POKHARA UNIVERSITY

Level: Bachelor | Semester: Spring | Year : 2021
Programme: BE | Full Marks: 100
Course: Engineering Mathematics II | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Prove that the lines $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ and $\frac{x-2}{3}=\frac{y-3}{4}=\frac{z-4}{5}$ are coplanar. Also Obtain the equation of the plane containing them. 7

OR

Define the shortest distance between the lines in space. Find the length and shortest distance between the lines $\frac{x}{2}=\frac{y}{-3}=\frac{z}{1}$ and $\frac{x-2}{3}=\frac{y-1}{-5}=\frac{z+3}{2}$

b) Find the equation of the sphere having circle $x^2+y^2+z^2=9$, $x-2y+2z=5$ as a great circle. Also determine its radius and center. 8

2. a) State and prove Euler's theorem for homogeneous function of three variables with degree n. 7

b) What do you mean by local maxima and minima? Find the dimension of rectangular box, open at the top, of maximum capacity whose surface is 432sq units 8

3. a) Evaluate the integral $\int_0^{4a}\int_{y^2/4a}^{y}\frac{x^2-y^2}{x^2+y^2}\,dxdy$ changing into polar co-ordinates. 7

b) Find the volume bounded by the z=0, the cylinder $x^2+y^2=1$ and the plane x+y+z=3. 8

4. a) What do you mean by Bornoulli's differential equation? Solve $\frac{dy}{dx}+\frac{1}{x}\sin 2y = x^3\cos^2 y$. 7

b) Find the general solution of $y''-4y'+4y=6+\frac{e^{2x}}{x}$, using variation of parameter method. 8

5. a) Solve $y''-4xy'+(4x^2-2)y=0$ by using power series method 7

OR

Define Legender's equation. Also derive the solution of Legender's equation. 8

b) Solve the initial value problem.

$y''-y'-2y=3e^{2x}$ given that $y(0)=0$ *and* $y'(0)=-2$

6. a) Define Laplace transform. Evaluate 8

i) $L(t^2\sin wt)$ ii) $L^{-1}\left(\frac{1}{s^2(s^2+w^2)}\right)$

b) Using Laplace Transform solve the initial value problem. 7

$y''-4y'+3y=e^{-t}$, $y(0)=y'(0)=1$

7. Attempt all questions:

a) What is the condition satisfied by the equation of a cone to have three mutually perpendicular generators. 2.5

b) If $F(s)=\frac{5s}{s^2-2s-3}$, find f(t). 2.5

c) Find the equation of sphere described on the join of (4,5,6) and (2,3,4) as a diameter. 2.5

d) If $u=e^{xyz}$, find $\frac{\partial^3 u}{\partial x\partial y\partial z}$. 2.5